

# Oppgave 1

$$S_b = 100 \text{ MVA} \quad V_b = 11 \text{ kV}$$

$$\text{Buss 1} \quad V_1 = 1.0 \angle 0^\circ$$

Jeg har brukt

$P = -0.5 \text{ pu}$  men skulle vært  $-0.3$

$$\text{Buss 2} \quad P_2 = -0.5 \text{ pu} \quad Q_2 = -0.2 \text{ pu}$$

$$Z_{12} = 0.1 + j0.5 \text{ pu}$$

$$V_2^{(0)} = 1 + j0$$

$$y = \frac{1}{Z_{12}} = \frac{1}{0.1 + j0.5} = \frac{0.1 - j0.5}{0.1^2 + 0.5^2}$$

$$= \frac{0.1 - j0.5}{0.26} = \underline{\underline{0.3846 - j1.9231}}$$

$$Y = \begin{bmatrix} y & -y \\ -y & y \end{bmatrix}$$

$$\frac{1}{Y_{22}} = 0.1 + j0.5$$

$$Y_{12} = Y_{21} = -y = -0.3846 + j1.9231$$

$$Y_{11} = Y_{22} = 0.3846 - j1.9231$$

a)

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$$V_2^{(1)} = \frac{1}{Y_{22}} \left( \frac{P_2 - jQ_2}{V_2^{(0)}} - Y_{21} \cdot V_1 \right)$$

$$\frac{P_2 - jQ_2}{V_2^{(0)}} = \frac{-0.5 + j0.2}{1} = \underline{\underline{-0.5 + j0.2}}$$

$$\begin{aligned} Y_{21} \cdot V_1 &= (-0.3846 + j1.9231) \cdot 1 \\ &= (-0.3846 + j1.9231) \end{aligned}$$

$$V_2^{(1)} = (0.1 + j0.5) \left( -0.5 + j0.2 + 0.3846 - j1.9231 \right)$$

$$= (0.1 + j0.5) (-0.1154 - j1.7231)$$

$$\underline{\underline{V_2^{(1)} = 0.85 - j0.2308}}$$

$$b) \quad V_2^{(2)} = \frac{1}{Y_{22}} \left( \frac{P - jQ}{(V_2^{(1)})^*} - Y_{21} \cdot V_1 \right)$$

$$V_2^{(1)*} = 0.85 + j0.2308$$

$$\frac{P - jQ}{V_2^{(1)*}} = \frac{-0.5 + j0.2}{0.85 + j0.2308}$$

$$= \frac{(-0.5 + j0.2)(0.85 - j0.2308)}{0.85^2 + 0.2308^2}$$

$$= \frac{-0.37885 + j0.2854}{0.7758}$$

$$\underline{\underline{= -0.4883 + j0.3679}}$$

$$Y_{21} \cdot V_1 = (-0.3846 + j1.9231)$$

$$V_2^{(2)} = (0.1 + j0.5) \left( \begin{array}{l} -0.4463 + j0.3679 - 0.3846 \\ + j1.9231 \end{array} \right)$$

$$= (0.1 + j0.5) (-0.1037 - j1.5552)$$

$$= \underline{\underline{0.7672 - j0.2074}}$$

C) 

$$V_2^{(2)} = 0.7672 + j0.2074$$

$$\frac{1}{Y_{22}} = 0.1 + j0.5$$

$$\frac{P_2 - jQ}{V_2^{(2)}} = \frac{-0.5 + j0.2}{0.7672 + j0.2074}$$

$$= \frac{(-0.5 + j0.2)(0.7672 - j0.2074)}{0.7672^2 + 0.2074^2}$$

$$= \frac{-0.34212 + j0.25714}{0.6316}$$

$$= \underline{\underline{-0.5417 + j0.4071}}$$

$$V_2^{(3)} = (0.1 + j0.5)(-0.5417 + j0.4071 + 0.3846 - j1.9231)$$

$$= (0.1 + j0.5)(-0.1571 - j1.516)$$

$$= \underline{\underline{0.7423 - j0.2301}}$$

$$V_2^{(3)*} = 0.7423 + j0.2301$$

$$\frac{P_2 - jQ}{V_2^{(3)}} = \frac{-0.5 + j0.2}{0.7423 + j0.2301}$$

$$= \frac{(-0.5 + j0.2)(0.7423 - j0.2301)}{0.7423^2 + 0.2301^2}$$

$$= \frac{-0.3251 + j0.2635}{0.6042} = \underline{\underline{-0.5381 + j0.4361}}$$

$$V_2^{(4)} = (0.1 + j0.5)(-0.5381 + j0.4361 + 0.3846 - j1.9231)$$

$$= (0.1 + j0.5)(-0.1535 - j1.4870)$$

$$= \underline{\underline{0.7262 - j0.2255}}$$

$$\underline{V_2^{(1)} = 0.85 - j0.2308}$$

$$\underline{V_2^{(2)} = 0.7672 - j0.2074}$$

$$\underline{V_2^{(3)} = 0.7423 - j0.2301}$$

$$\underline{V_2^{(4)} = 0.7262 - j0.2255}$$

d)

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$$|V_2| = \sqrt{0.7282^2 + 0.2255^2}$$

$$= 0.7623 \text{ pu}$$

$$\angle = \tan^{-1} \left( \frac{-0.2255}{0.7282} \right) = \underline{\underline{\sim -17.2^\circ}}$$

# oppgave 2

a)

```
1  #-*- coding: utf-8 -*-
2  """
3  Created on Thu Sep 24 15:15:57 2026
4
5  @author: Aksel
6  """
7  import pandapower as pp
8
9  net = pp.create_empty_network()
10
11  # Busser
12  b1 = pp.create_bus(net, vn_kv=11, name="Bus 1")
13  b2 = pp.create_bus(net, vn_kv=11, name="Bus 2")
14
15  # Slack
16  pp.create_ext_grid(
17      net,
18      bus=b1,
19      vm_pu=1.0
20  )
21
22  # Impedans
23  pp.create_impedance(
24      net,
25      from_bus=b1,
26      to_bus=b2,
27      rft_pu=0.1,
28      xft_pu=0.5,
29      sn_mva=100
30  )
31
32  # Last
33  pp.create_load(
34      net,
35      bus=b2,
36      p_mw=50,
37      q_mvar=20
38  )
39
40  # Lastflyt
41  pp.runpp(net, algorithm="nr")
42
43  print(net.res_bus)
```

← jeg endret denne siden  
jeg brukte  
 $p = -0.6$  pu

b)

```
In [6]: runfile('C:/Uis/ELK330/Lab/Lab 1/Spyder.py', wdir='C:/Uis/
ELK330/Lab/Lab 1')
numba cannot be imported and numba functions are disabled.
Probably the execution is slow.
Please install numba to gain a massive speedup.
(or if you prefer slow execution, set the flag numba=False to avoid this
warning!)

   vm_pu  va_degree      p_mw      q_mvar
0  1.000000  0.000000 -55.11441 -45.572049
1  0.753011 -17.784666  50.00000  20.000000
```

c)

Gauss :  $0,7623 \angle -17,2^\circ$

Python :  $0,753 \angle -17,78$

de er ikke like, men forskjellen  
tror jeg kommer fra avrunding

### oppgave 3

a) Løs lastflyten

b) Noter  $V_2$  og  $S_2$

c) Sammenlign

- Gauss

- Newton

- Panda power

d)

1.

Lasten på Buss 2 trækker  $P$  og  $Q$

Det er spændingsfald på linjen.

Det giver at spændingen ved Buss 2 er  
lavere end slack-bussen.

2.

Spændingsvinkel på Buss 2 er  $\approx -17^\circ$

i forhold til slack-buss. Det viser at  
effekten flytter fra slack-buss til Buss 2

3.

Ligneimpedansen påvirker spenningsfallet med at når impedansen øker så blir spenningsfall. Det gjør at spenningen ved Buss 2 blir lavere.

4.

Når impedansen dobles så blir det større spenningsfall.

Det gjør at spenningen på Buss 2 blir mindre og vinkelen blir også lavere.